

B.E. ELECTRICAL ENGINEERING
SECOND YEAR
SECOND SEMESTER EXAM 2025
FIELD THEORY
PART-II

Solved Answer Script in Academic Article Style

Time: 3 hours

Use separate answer-script for each part

Full Marks: 100

50 marks for each part

Answer any three questions. Two marks are for neatness. All symbols have their usual significance.

Final result / exam point

This solution writes **all questions** of Part-II so that a student can choose any required three in the semester exam. Each answer follows:

Question heading → answer-building plan → diagram → derivation → final result

1. (a) What is the fundamental definition of curl of a vector field? Derive the expression for curl of vector field $\mathbf{M}$, written as $\nabla \times \mathbf{M}$ where ∇ is the vector differential operator in Cartesian co-ordinates.

Short answer-building plan

First write the fundamental definition of curl using circulation per unit area. Then take a very small rectangular loop in the xy -plane. Find circulation of $\mathbf{M}$ around the four sides. Divide by area $dx dy$ to get the z -component. Finally write the complete Cartesian expression using cyclic symmetry.

Detailed semester-style answer

Definition of curl:

Curl of a vector field gives the local rotational tendency or circulation tendency of the field at a point.

For a vector field $\mathbf{M}$, the component of curl along unit normal $\hat{\mathbf{n}}$ is defined as

$$(\nabla \times \mathbf{M}) \cdot \hat{\mathbf{n}} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \mathbf{M} \cdot d\mathbf{l}$$

where C is the boundary of the small surface ΔS .

Let

$$\mathbf{M} = M_x \hat{\mathbf{x}} + M_y \hat{\mathbf{y}} + M_z \hat{\mathbf{z}}.$$

To derive the z -component of curl, consider a small rectangular path in the xy -plane.

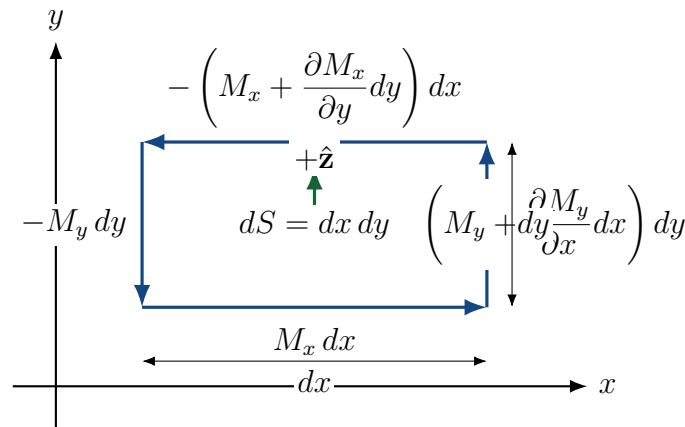


Figure 1: Small rectangular path in the xy -plane for deriving the z -component of curl.

Detailed semester-style answer

Now calculate the circulation:

$$\oint_C \mathbf{M} \cdot d\mathbf{l} = \text{sum of line integrals over four sides.}$$

For side 1:

$$\int_1 \mathbf{M} \cdot d\mathbf{l} = M_x dx.$$

For side 2:

$$\int_2 \mathbf{M} \cdot d\mathbf{l} = \left(M_y + \frac{\partial M_y}{\partial x} dx \right) dy.$$

For side 3:

$$\int_3 \mathbf{M} \cdot d\mathbf{l} = - \left(M_x + \frac{\partial M_x}{\partial y} dy \right) dx.$$

For side 4:

$$\int_4 \mathbf{M} \cdot d\mathbf{l} = -M_y dy.$$

Therefore,

$$\begin{aligned} \oint_C \mathbf{M} \cdot d\mathbf{l} &= M_x dx + \left(M_y + \frac{\partial M_y}{\partial x} dx \right) dy \\ &\quad - \left(M_x + \frac{\partial M_x}{\partial y} dy \right) dx - M_y dy. \end{aligned}$$

After cancellation:

$$\oint_C \mathbf{M} \cdot d\mathbf{l} = \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} \right) dx dy.$$

Since the area is

$$\Delta S = dx dy,$$

we get

$$(\nabla \times \mathbf{M})_z = \frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}$$

Similarly,

$$(\nabla \times \mathbf{M})_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z}$$

and

$$(\nabla \times \mathbf{M})_y = \frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x}$$

Final result / exam point

Hence, in Cartesian co-ordinates,

$$\nabla \times \mathbf{M} = \left(\frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y} \right) \hat{\mathbf{z}}$$

or,

$$\nabla \times \mathbf{M} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M_x & M_y & M_z \end{vmatrix}$$

1. (b) Establish the Stoke's Theorem and $\nabla \times \mathbf{H} = \mathbf{J}$.

Short answer-building plan

First state Stoke's theorem as the relation between line integral and surface integral of curl. Then explain it by dividing an open surface into many small surface elements. After that apply Ampere's circuital law and convert it into differential form using Stoke's theorem.

Detailed semester-style answer

Stoke's theorem:

Stoke's theorem states that the line integral of a vector field around a closed boundary is equal to the surface integral of the curl of that vector field over the open surface bounded by that boundary.

Mathematically,

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Here,

- C is the closed boundary curve,
- S is any open surface bounded by C ,
- $d\mathbf{S}$ is the vector surface element.

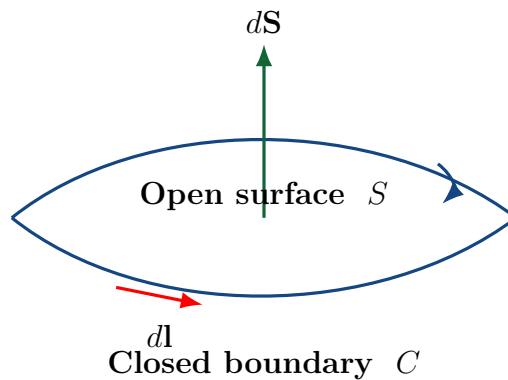


Figure 2: Stoke's theorem connects boundary circulation with curl over the surface.

Detailed semester-style answer

For a very small surface element,

$$(\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

represents the small circulation around that element.

If we divide the whole surface into many small elements and add the circulations, then all internal boundary line integrals cancel because they are taken in opposite directions.

Only the outer boundary remains. Hence,

$$\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_C \mathbf{A} \cdot d\mathbf{l}$$

This establishes Stoke's theorem.

Now establish $\nabla \times \mathbf{H} = \mathbf{J}$.

Ampere's circuital law states that

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

The enclosed current can be written in terms of current density:

$$I_{\text{enc}} = \iint_S \mathbf{J} \cdot d\mathbf{S}$$

Therefore,

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

By Stoke's theorem,

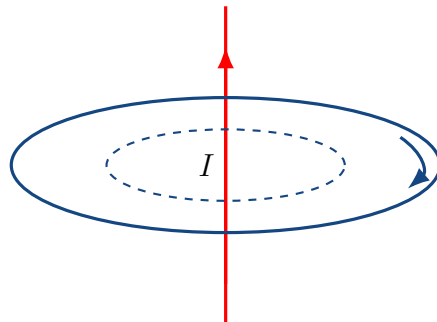
$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}.$$

Thus,

$$\iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

Since this is true for any arbitrary surface S ,

$$\nabla \times \mathbf{H} = \mathbf{J}$$



$\mathbf{H}$ circulates around current

Current density $\mathbf{J}$ pierces the surface

Figure 3: Physical view of $\nabla \times \mathbf{H} = \mathbf{J}$. Current density produces circulation of magnetic field intensity.

Final result / exam point

$$\nabla \times \mathbf{H} = \mathbf{J}$$

means current density is the source of circulation of magnetic field intensity.

2. (a) Derive the expression for Divergence of vector field $\mathbf{B}$ in Cartesian co-ordinates using fundamental definition.

Short answer-building plan

Start from the definition of divergence as outward flux per unit volume. Take a small rectangular box of dimensions dx , dy , and dz . Find net flux through the six faces. Divide by the volume to get the Cartesian expression.

Detailed semester-style answer

Divergence of a vector field gives the net outward flux per unit volume at a point.

For vector field $\mathbf{B}$,

$$\nabla \cdot \mathbf{B} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oint_S \mathbf{B} \cdot d\mathbf{S}$$

Let

$$\mathbf{B} = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}.$$

Consider a small rectangular volume:

$$\Delta V = dx \, dy \, dz.$$

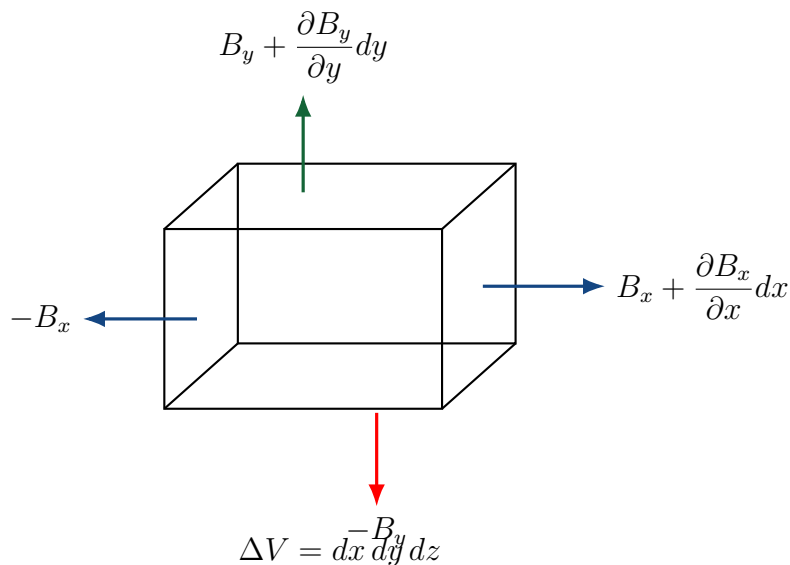


Figure 4: Small rectangular volume used to derive divergence in Cartesian coordinates.

Detailed semester-style answer

Net flux through the two faces perpendicular to the x -axis is

$$\left[\left(B_x + \frac{\partial B_x}{\partial x} dx \right) - B_x \right] dy \, dz = \frac{\partial B_x}{\partial x} dx \, dy \, dz.$$

Net flux through the two faces perpendicular to the y -axis is

$$\left[\left(B_y + \frac{\partial B_y}{\partial y} dy \right) - B_y \right] dx dz = \frac{\partial B_y}{\partial y} dx dy dz.$$

Net flux through the two faces perpendicular to the z -axis is

$$\left[\left(B_z + \frac{\partial B_z}{\partial z} dz \right) - B_z \right] dx dy = \frac{\partial B_z}{\partial z} dx dy dz.$$

Therefore, total outward flux is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right) dx dy dz.$$

Dividing by volume,

$$\Delta V = dx dy dz,$$

we get

$$\nabla \cdot \mathbf{B} = \frac{\oint_S \mathbf{B} \cdot d\mathbf{S}}{dx dy dz}.$$

Final result / exam point

$$\boxed{\nabla \cdot \mathbf{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}}$$

This is the expression for divergence of $\mathbf{B}$ in Cartesian co-ordinates.

2. (b) Explain the physical significance of $\nabla \cdot \mathbf{J} = 0$.

Short answer-building plan

Write what current density means. Then use the continuity equation. Show that $\nabla \cdot \mathbf{J} = 0$ means no charge accumulation. Finally connect it with steady current and Kirchhoff's current law.

Detailed semester-style answer

Current density $\mathbf{J}$ represents current flowing per unit area. Its divergence,

$$\nabla \cdot \mathbf{J},$$

represents the net current flowing out per unit volume.

The continuity equation is

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

where ρ_v is volume charge density.

If

$$\nabla \cdot \mathbf{J} = 0,$$

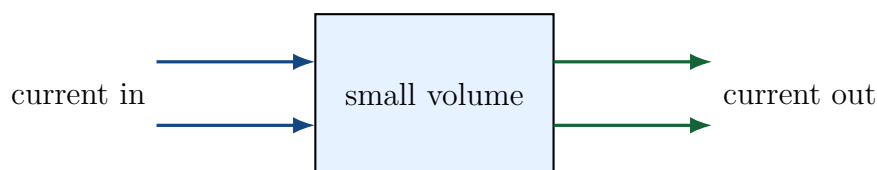
then

$$-\frac{\partial \rho_v}{\partial t} = 0.$$

Hence,

$$\frac{\partial \rho_v}{\partial t} = 0$$

So, charge density is not changing with time.



$$\nabla \cdot \mathbf{J} = 0 \Rightarrow \text{no charge accumulation}$$

Figure 5: For $\nabla \cdot \mathbf{J} = 0$, current entering equals current leaving.

Final result / exam point

Physical significance of

$$\nabla \cdot \mathbf{J} = 0$$

is:

- no net current source or sink exists inside the small volume,
- current entering equals current leaving,
- no charge accumulation occurs,

- it represents steady current condition,
- it is the field form of Kirchhoff's current law.

2. (c) Explain that $\nabla \cdot \mathbf{B} = 0$.

Short answer-building plan

State Gauss's law for magnetism. Explain that magnetic flux lines never start or end. They form closed loops. Hence the net magnetic flux through any closed surface is zero.

Detailed semester-style answer

For magnetic flux density $\mathbf{B}$, Gauss's law for magnetism is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

Using divergence theorem,

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{B} dV.$$

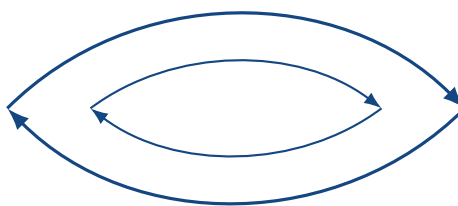
So,

$$\iiint_V \nabla \cdot \mathbf{B} dV = 0.$$

Since the volume is arbitrary,

$$\nabla \cdot \mathbf{B} = 0$$

$\mathbf{B}$ field lines are closed



No isolated magnetic pole exists

Figure 6: Magnetic flux lines are closed loops, so their divergence is zero.

Final result / exam point

$$\nabla \cdot \mathbf{B} = 0$$

means:

- magnetic flux has no source or sink,
- isolated magnetic monopoles do not exist,
- magnetic field lines are closed loops,

- net magnetic flux through a closed surface is zero.

3. (a) Establish $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$ and show the pictorial view of this relation.

Short answer-building plan

Begin from Ampere's law for steady current. Then explain why it fails for time-varying electric field. Introduce displacement current density $\partial \mathbf{D} / \partial t$. Then write Ampere-Maxwell law in integral form and convert it into differential form.

Detailed semester-style answer

For steady current, Ampere's law is

$$\nabla \times \mathbf{H} = \mathbf{J}$$

But for time-varying fields, conduction current alone is not sufficient. Maxwell introduced displacement current density:

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

Therefore, total current density is

$$\mathbf{J}_{total} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Hence, the complete Ampere-Maxwell law becomes

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

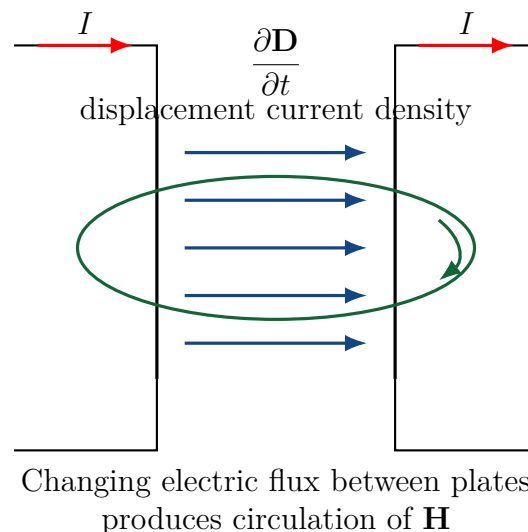


Figure 7: Pictorial view of Ampere-Maxwell law.

Detailed semester-style answer

The integral form of Ampere-Maxwell law is

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$$

Using Stoke's theorem:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}.$$

Therefore,

$$\iint_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \iint_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}.$$

Since this is true for any surface,

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Final result / exam point

This equation means that magnetic field circulation is produced by both:

conduction current density $\mathbf{J}$

and

displacement current density $\frac{\partial \mathbf{D}}{\partial t}$

3. (b) Derive $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ and show the pictorial view of this relation.

Short answer-building plan

Start from Faraday's law of electromagnetic induction in integral form. Then express magnetic flux through a surface. Use Stoke's theorem on the electric field line integral. Finally compare both surface integrals and get the differential form.

Detailed semester-style answer

Faraday's law states that a time-varying magnetic flux produces an induced electric field. The integral form is

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{S}$$

If the surface is stationary, then

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}$$

Using Stoke's theorem:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S}.$$

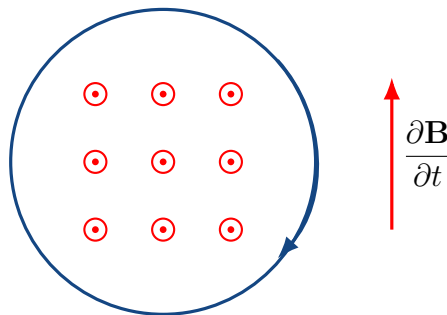
Thus,

$$\iint_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S}.$$

Since this is true for any surface,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Changing magnetic field



Induced circulating electric field $\mathbf{E}$

Figure 8: Pictorial view of Faraday's law. Changing $\mathbf{B}$ produces circulating $\mathbf{E}$.

Final result / exam point

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

The negative sign represents Lenz's law. The induced electric field opposes the cause producing it.

4. (a) Derive Electromagnetic Wave equations from Maxwell's equations.

Short answer-building plan

Write Maxwell's curl equations in a homogeneous medium. Use material relations $\mathbf{D} = \epsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$, and $\mathbf{J} = \sigma\mathbf{E}$. Take curl of Faraday's law for the electric field wave equation. Take curl of Ampere-Maxwell law for the magnetic field wave equation.

Detailed semester-style answer

For a homogeneous conducting medium:

$$\mathbf{D} = \epsilon\mathbf{E}, \quad \mathbf{B} = \mu\mathbf{H}, \quad \mathbf{J} = \sigma\mathbf{E}.$$

Maxwell's curl equations are

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

and

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma\mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}$$

Wave equation for electric field:

Take curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}).$$

Substitute

$$\nabla \times \mathbf{H} = \sigma\mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}.$$

Then

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} \left(\sigma\mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t} \right).$$

So,

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Using vector identity:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}.$$

In source-free region,

$$\nabla \cdot \mathbf{E} = 0.$$

Therefore,

$$-\nabla^2 \mathbf{E} = -\mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Hence,

$$\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Wave equation for magnetic field:

Similarly,

$$\nabla^2 \mathbf{H} = \mu\sigma \frac{\partial \mathbf{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

Final result / exam point

The electromagnetic wave equations are:

$$\nabla^2 \mathbf{E} = \mu\sigma \frac{\partial \mathbf{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

and

$$\nabla^2 \mathbf{H} = \mu\sigma \frac{\partial \mathbf{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

4. (b) Obtain E.M. wave equations in free space and then find an analytical solution of the wave equation of Magnetic Field ($\mathbf{H}$) in free space considering it as a plane wave and also draw the wave propagation.

Short answer-building plan

First put free-space conditions $\sigma = 0$, $\mu = \mu_0$, and $\epsilon = \epsilon_0$. Then write the free-space wave equation for $\mathbf{H}$. Assume plane wave propagation along $+x$ direction. Reduce the vector wave equation to one scalar equation for $H_z(x, t)$. Solve it using sinusoidal travelling wave form.

Detailed semester-style answer

In free space:

$$\sigma = 0, \quad \mu = \mu_0, \quad \epsilon = \epsilon_0$$

Therefore, the wave equations become:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

and

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

Let the plane wave travel in the $+x$ direction. For a plane wave, field quantities vary only with x and t .

Let the magnetic field be along z direction:

$$\mathbf{H} = H_z(x, t) \hat{\mathbf{z}}$$

Then

$$\nabla^2 \mathbf{H} = \frac{\partial^2 H_z}{\partial x^2} \hat{\mathbf{z}}.$$

So the wave equation becomes:

$$\frac{\partial^2 H_z}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 H_z}{\partial t^2}$$

Analytical solution:

A sinusoidal travelling wave solution is

$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

where

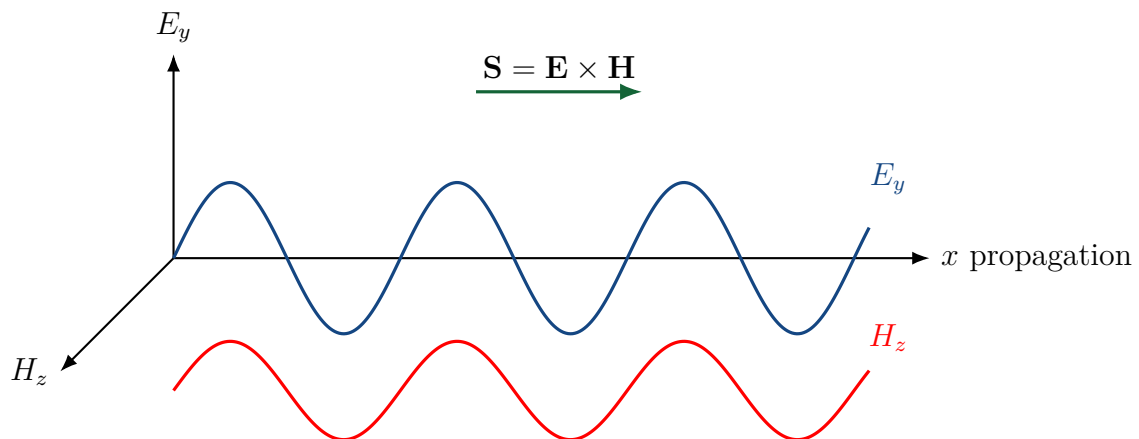
$$\beta = \omega \sqrt{\mu_0 \epsilon_0}$$

The velocity of the wave is

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

In free space,

$$v = c \approx 3 \times 10^8 \text{ m/s}$$



Plane EM wave: $\mathbf{E}$, $\mathbf{H}$ and direction of propagation are mutually perpendicular

Figure 9: Wave propagation in free space. Electric and magnetic fields are transverse to the direction of propagation.

Final result / exam point

For plane wave propagation along $+x$ direction:

$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

where

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}$$

and

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s}$$

5. Write short notes on any four:

Short answer-building plan

The paper asks for any four, but all five short notes are written here. In the exam, choose the four you can present most confidently with formula, meaning, and one neat diagram where useful.

5. (a) Physical significance of Curl;

Detailed semester-style answer

Curl represents the local rotational tendency of a vector field.

If a small paddle wheel is placed in a vector field, then curl tells whether the wheel will rotate and about which axis.

Mathematically,

$$\nabla \times \mathbf{A}$$

represents the curl of vector field $\mathbf{A}$.

If

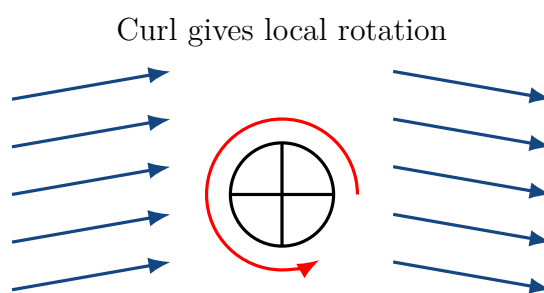
$$\nabla \times \mathbf{A} = 0,$$

then the field is irrotational.

If

$$\nabla \times \mathbf{A} \neq 0,$$

then the field has circulation or rotation.



$$\nabla \times \mathbf{A} \neq 0 \text{ means rotational field}$$

Figure 10: Physical meaning of curl using paddle-wheel idea.

Final result / exam point

In electromagnetics,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

means current density produces circulation of magnetic field intensity.

5. (b) Poynting Vector;

Detailed semester-style answer

Poynting vector represents the rate of electromagnetic energy flow per unit area.
It is defined as

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

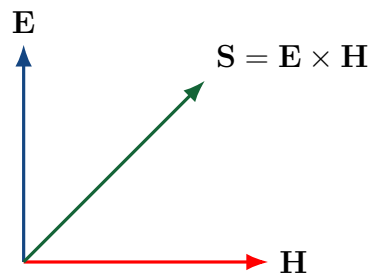
The direction of $\mathbf{S}$ is perpendicular to both $\mathbf{E}$ and $\mathbf{H}$.

The unit of Poynting vector is

$$\text{W/m}^2$$

The total electromagnetic power crossing a surface is

$$P = \iint_S \mathbf{S} \cdot d\mathbf{S}$$



Direction of energy flow

Figure 11: Poynting vector direction.

Final result / exam point

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

It gives both magnitude and direction of electromagnetic power flow per unit area.

5. (c) EM Plane wave velocity in free space;

Detailed semester-style answer

The wave equation in free space is

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Comparing with standard wave equation,

$$\nabla^2 \mathbf{E} = \frac{1}{v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2},$$

we get

$$\frac{1}{v^2} = \mu_0 \epsilon_0.$$

Therefore,

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Using

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

and

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m},$$

we get

$$v \approx 3 \times 10^8 \text{ m/s}$$

Final result / exam point

The velocity of an EM plane wave in free space is equal to the velocity of light:

$$c = 3 \times 10^8 \text{ m/s}$$

5. (d) Displacement current;

Detailed semester-style answer

Displacement current is the current-like effect produced by a time-varying electric flux density. Displacement current density is defined as

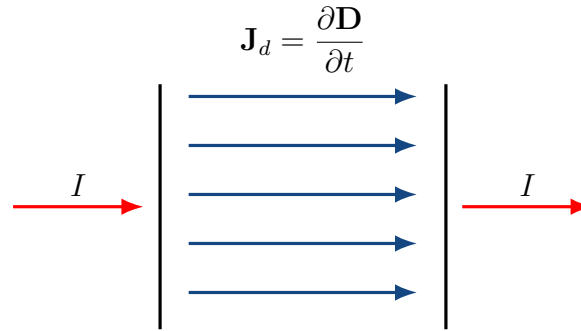
$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

The displacement current through a surface is

$$I_d = \iint_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S}$$

It was introduced by Maxwell to modify Ampere's law for time-varying fields:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$



Displacement current exists between capacitor plates

Figure 12: Displacement current between capacitor plates.

Final result / exam point

Displacement current makes current continuity possible in time-varying electric fields and completes Maxwell's equations.

5. (e) Divergence of the curl of any vector field is always zero.

Short answer-building plan

Write the statement first. Then prove it using Cartesian expansion of $\nabla \times \mathbf{A}$. Finally show that all mixed partial derivative terms cancel.

Detailed semester-style answer

For any vector field $\mathbf{A}$,

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

Let

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}.$$

Then

$$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{\mathbf{z}}.$$

Taking divergence,

$$\begin{aligned} \nabla \cdot (\nabla \times \mathbf{A}) &= \frac{\partial}{\partial x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \\ &\quad + \frac{\partial}{\partial y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ &\quad + \frac{\partial}{\partial z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right). \end{aligned}$$

Expanding,

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{A}) = & \frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} \\ & - \frac{\partial^2 A_z}{\partial y \partial x} + \frac{\partial^2 A_y}{\partial z \partial x} - \frac{\partial^2 A_x}{\partial z \partial y}.\end{aligned}$$

For continuous fields, mixed partial derivatives are equal:

$$\frac{\partial^2 A_z}{\partial x \partial y} = \frac{\partial^2 A_z}{\partial y \partial x},$$

$$\frac{\partial^2 A_y}{\partial x \partial z} = \frac{\partial^2 A_y}{\partial z \partial x},$$

$$\frac{\partial^2 A_x}{\partial y \partial z} = \frac{\partial^2 A_x}{\partial z \partial y}.$$

Therefore all terms cancel.

$$\boxed{\nabla \cdot (\nabla \times \mathbf{A}) = 0}$$

Final result / exam point

The divergence of curl is always zero because curl represents circulation, and pure circulation has no source or sink.